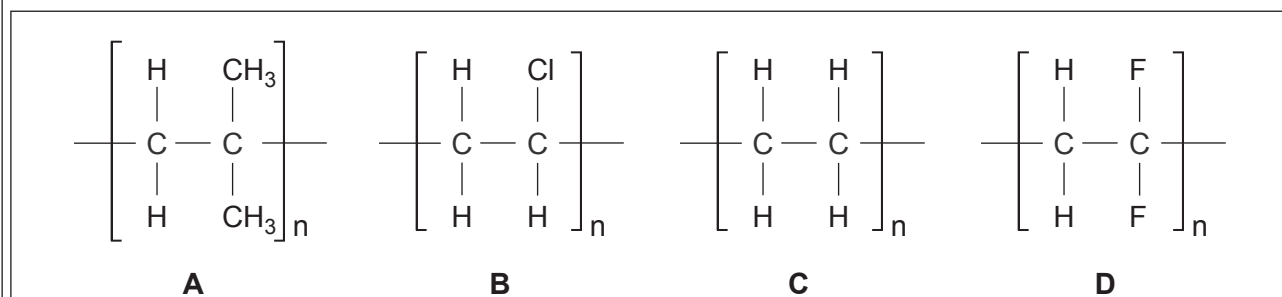


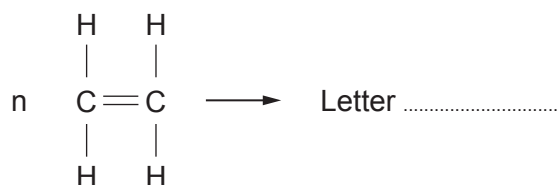
3. (a) Complete the table.

[2]

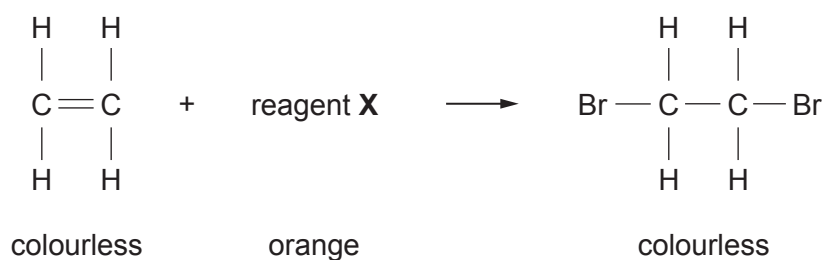
Name	Molecular formula	Structural formula
ethane	C_2H_6	<pre> H H H — C — C — H H H </pre>
propane	C_3H_8	
butane		<pre> H H H H H — C — C — C — C — H H H H H </pre>

(b) The box contains four polymers **A**, **B**, **C** and **D**.Give the **letter** of the polymer formed during the polymerisation of ethene.

[1]



(c) The equation shows the reaction taking place when testing for a $C=C$ bond.



bromine water potassium dichromate universal indicator barium chloride

Choose the name of reagent **X** from the box above.

..... [1]

(d) The equation shows the fermentation of glucose.



(i) Underline the chemical name for $\text{C}_2\text{H}_5\text{OH}$. [1]

sugar biofuel ethanol alcohol

(ii) Calculate the relative molecular mass, M_r , of $\text{C}_2\text{H}_5\text{OH}$. [2]

$$A_r(\text{H}) = 1 \quad A_r(\text{O}) = 16 \quad A_r(\text{C}) = 12$$

$$M_r = \dots\dots\dots$$

